MSA Technologies

Official Product Development Documentation

Online Restaurant - Complete Restaurant Management & Automation System



Prepared for the four-member product team: Ali, Ishtiaq, Sanwal, and Asad

Document Status: Internal Team Planning Document | Version: 1.0 | Date: May 2026

1. Document Control

Company	MSA Technologies
Product Name	Online Restaurant
Document Type	Official Product Planning and Team Documentation
Team Size	4 members
Team Members	Ali, Ishtiaq, Sanwal, and Asad
Purpose	To guide the team in building the product from scratch as one collaborative unit.
Market Focus	Pakistan and international restaurant industry

2. Project Vision

MSA Technologies will build Online Restaurant as a complete digital platform for the restaurant industry. The product will help restaurants manage websites, apps, CRM, WhatsApp orders, customer calls, menus, dashboards, delivery workflows, reporting, and daily operations from one smart system.

The platform must be practical for restaurants in Pakistan while also being scalable for international restaurant businesses. The team will build the product from scratch and will work together as one unit. Internal responsibilities can be adjusted according to skills, availability, and project requirements.

3. Main Objectives

- Create a complete restaurant technology product that can be sold as a business solution.
- Enable restaurants to receive and manage online orders through website, app, and WhatsApp.
- Reduce missed messages, missed calls, and missed orders using automation.
- Provide restaurant owners with a dashboard for sales, customers, orders, menus, and reports.
- Support local Pakistani restaurant operations and international restaurant standards.
- Develop a strong MVP first, test it with real restaurants, then expand into advanced features.

4. Product Scope

The product will include the following major modules. These modules should be planned as one connected system, not as separate disconnected tools.

Module	Purpose
Restaurant Website	Professional restaurant website with online menu, order placement, offers, reviews, location, and order tracking.
Mobile App	Customer app for browsing menus, ordering food, tracking orders, receiving offers, and reordering.
Admin Dashboard	Management dashboard for orders, menus, customers, branches, staff, delivery status, sales, and reporting.
CRM System	Customer profiles, order history, preferences, complaints, feedback, loyalty points, and marketing campaigns.
WhatsApp Order Automation	Automated menu sharing, order taking, address collection, confirmation, delivery updates, and human handover.
AI Phone Agent	Voice assistant to answer calls, take basic orders, provide menu details, confirm information, and reduce missed calls.
Digital Menu System	Menu categories, prices, images, item availability, deals, combos, QR menu, and branch-wise menu control.
Order & Delivery Management	Kitchen order status, pickup, delivery, rider assignment, cancellation, completion, and order history.
Reports & Analytics	Daily sales, monthly sales, revenue, total orders, top items, repeat customers, delivery performance, and campaign reports.

5. Target Users

- Restaurant owners and management teams
- Restaurant branch managers
- Kitchen and order handling staff
- Delivery staff or rider teams
- Customers ordering through website, mobile app, or WhatsApp
- Marketing and customer support teams

6. Development Approach

The product should be developed in phases. The first goal is to launch a strong MVP that solves the most important restaurant problems. After pilot testing, the team can expand the system with advanced automation, integrations, and international features.

1. Start with clear research and requirements.
2. Design user journeys before development.
3. Build the backend, database, website, dashboard, app, CRM, and automation in a structured order.
4. Test every core workflow before pilot launch.
5. Launch pilot with selected restaurants and collect real feedback.
6. Improve the product and prepare for commercial launch.

7. Development Phases and Timeline

Phase	Estimated Duration	Main Output
1. Market Research & Requirement Planning	1-2 weeks	Requirement document, market needs, user roles, and MVP plan.
2. UI/UX Design	2 weeks	Designs for website, dashboard, app, CRM, menu, and WhatsApp flow.
3. Backend & Database Development	3-4 weeks	Authentication, database, APIs, roles, restaurant profile, menu, orders, CRM, and reports.
4. Website & Admin Dashboard	3-4 weeks	Restaurant website, online ordering, dashboard, menu management, order management, and reports.
5. Mobile App Development	4-5 weeks	Customer app with menu, cart, checkout, order tracking, profile, and notifications.
6. WhatsApp Automation & AI Phone Agent	3-4 weeks	WhatsApp order flow, automated replies, order confirmation, admin alerts, and AI call workflow.
7. CRM, Reports & Business Tools	2-3 weeks	Customer database, order history, loyalty tools, campaigns, sales reports, and insights.
8. Testing & Quality Assurance	2 weeks	Stable product with tested workflows, security checks, and performance improvements.
9. Pilot Launch	2-4 weeks	Real restaurant testing, user feedback, bug fixes, improvements, and pricing model.
10. Commercial Launch	After pilot	Official launch, demos, onboarding, subscription packages, and support process.

8. MVP Scope

The MVP must include the minimum features required to deliver real value to restaurants and prove the product in the market.

- Restaurant website with online menu and online ordering
- Digital menu management with categories, prices, images, availability, deals, and combos
- Admin dashboard for restaurant owners and staff

- Order status management from new order to completed order
- Customer database and basic CRM
- WhatsApp order automation for menu sharing and order confirmation
- Basic sales and order reports
- Basic customer mobile app or mobile-friendly web ordering system
- Restaurant profile, branch setup, and role-based access

9. Advanced Features for Future Versions

- AI phone agent for calls and voice-based order handling
- Full loyalty and rewards system
- Multi-branch and franchise management
- Rider tracking and delivery route management
- Payment gateway integrations for local and international markets
- Inventory management and stock alerts
- POS integration
- Advanced analytics and predictive reporting
- Multi-language and multi-currency support
- International tax, delivery, and compliance settings

10. Suggested Technology Structure

Area	Suggested Structure
Frontend	Restaurant website, customer web ordering system, and admin dashboard.
Mobile App	Android-first approach, with option for cross-platform app and iOS later.
Backend	Secure APIs, authentication, role-based access, restaurant database, order database, CRM database, and reporting system.
Database	Structured database for users, restaurants, branches, menus, orders, payments, customers, campaigns, and reports.
Integrations	WhatsApp Business API, payment gateways, SMS gateway, email system, Google Maps, push notifications, and AI voice/call system.
Security	Role-based access, secure login, data validation, backups, audit logs, and protection of customer information.

11. Recommended Team Workflow

The four-member team should work together and adjust responsibilities internally based on skill, experience, and availability. The product should be managed through clear planning, task tracking, and regular review.

- Weekly planning meeting to review completed work, pending work, blockers, bugs, and next targets.
- Daily progress update from each team member covering completed work, next task, and required support.
- Shared task board for features, bugs, deadlines, testing status, deployment status, and feedback.
- Code review and testing before merging important changes.
- Product demo at the end of each major phase.
- Written record of all major decisions, feature changes, and client feedback.

12. Initial Backlog

Priority	Feature Area	Initial Task
High	Restaurant Setup	Create restaurant profile, branch setup, operating hours, contact details, and location.
High	Menu	Add categories, items, prices, images, descriptions, availability, deals, and combos.
High	Ordering	Build cart, checkout, order confirmation, and order status flow.
High	Dashboard	Create order list, order detail view, menu management, and basic reports.
High	WhatsApp	Create menu sharing, order capture, customer confirmation, and admin notification flow.
Medium	CRM	Create customer profiles, order history, customer notes, and feedback management.
Medium	Reports	Create daily sales, total orders, revenue, top items, and repeat customer reports.
Medium	Mobile App	Create menu browsing, cart, checkout, order tracking, customer profile, and notifications.
Future	AI Phone Agent	Design voice call workflow, call scripts, fallback rules, and order summary.
Future	Integrations	Plan payment gateway, SMS, email, Google Maps, POS, and inventory integrations.

13. Testing Checklist

- Restaurant website opens correctly on desktop and mobile.
- Menu items, prices, images, availability, deals, and categories update correctly.
- Customer can place an order from website or app without confusion.
- Admin receives and manages order status correctly.
- WhatsApp automation sends accurate menu, confirmation, and delivery updates.
- CRM stores customer information and order history correctly.
- Reports calculate revenue, orders, and top items accurately.
- User roles and permissions are secure.
- System handles slow internet and common user mistakes.
- Pilot restaurant staff can understand and use the system without heavy training.

14. Product Success Goals

- No missed messages, calls, or orders for restaurants using the system.
- Faster order processing and improved customer response time.
- Better restaurant online presence through website, app, and WhatsApp.
- Higher customer retention through CRM, loyalty, and campaigns.
- Clear reporting for sales, revenue, items, customers, and delivery performance.
- A scalable product that can be sold in Pakistan and expanded internationally.

15. Risk Management

Risk	Mitigation Plan
Unclear product scope	Lock MVP scope before development and move extra features to future versions.
Integration delays	Start with basic manual workflows, then add integrations step by step.
Restaurant staff adoption issues	Keep the dashboard simple, run pilot training, and collect staff feedback.
WhatsApp API or payment gateway complexity	Plan early, test with sandbox accounts, and keep manual fallback options.
Performance issues during order load	Design scalable backend, optimize database queries, and test under realistic order volume.
Feature overload	Prioritize restaurant pain points and avoid building unnecessary features in the MVP.

16. Recommended Launch Strategy

7. Build MVP focused on website ordering, digital menu, admin dashboard, WhatsApp automation, CRM, and basic reports.
8. Select 2-3 restaurants for pilot testing in Pakistan.
9. Collect feedback from owners, managers, staff, and customers.
10. Improve product based on real usage, not assumptions.
11. Prepare pricing packages such as monthly subscription, setup fee, and premium automation add-ons.
12. Launch with demos, restaurant visits, social media campaigns, and referral partnerships.

17. Final Recommendation

The four-member team should begin with a focused MVP and work together from scratch as one product team. The first development priority should be restaurant website, digital menu, online ordering, admin dashboard, WhatsApp automation, CRM, and basic reports. After pilot testing, the product should expand into mobile app improvements, AI phone agent, payments, rider tracking, inventory, POS integration, and international support.

This structured approach will help MSA Technologies build a professional, market-ready restaurant product that can serve local restaurants in Pakistan and scale to international restaurant businesses.